Project Design Phase-I
ProposedSolutionTemplate

Date	09 March 2026
ProjectName	UCab – Cab Booking Web Application

ProposedSolutionTemplate:

Projectteamshallfillthefollowinginformationinproposedsolutiontemplate.

S.No.	Parameter	Description
1.	ProblemStatement(Problemto be solved)	Urban transportation often suffers from inefficiency, long waiting times, and lack of transparency in pricing and driver availability. Traditional taxi services usually require customers to call booking centers or physically search for taxis, which can be time- consuming and unreliable. In many cities, passengers struggle to find available cabs during peak hours, while drivers also face difficulties managing bookings efficiently. There is a need for a centralized digital platform that allows users to easily search for available cabs, estimate fares, book rides, and manage bookings online. At the same time, administrators and fleet owners require a system that can manage vehicles, drivers, and bookings efficiently. UCab aims to solve these problems by providing a web-based cab booking platform that connects users with available vehicles and enables efficient ride management through a modern digital interface.
2.	Idea/Solutiondescription	UCab is a MERN-stack based web application designed to simplify the cab booking process. The platform allows users to register, browse available cabs, estimate fares, and book rides directly from a web interface. The system consists of two main modules: User Module: • Register and login securely using JWT authentication • Browse available cabs filtered by type (Mini, Sedan, SUV, Luxury) • View cab details including driver information and pricing • Book rides by entering pickup and destination locations • View estimated fare before confirming booking • Track booking history and cancel rides Admin Module: • Admin dashboard displaying system statistics • Manage users (view, edit, activate/deactivate) • Manage fleet of cabs (add, update, delete) • Upload cab images • View and update booking statuses The backend is implemented using Node.js and Express.js, the frontend uses React.js, and MongoDB stores all application data.

3.	Novelty/Uniqueness	The UCab system offers several unique features that improve the cab booking experience: • Role- based access control for users and administrators • Real- time booking management and status tracking • Dynamic fare estimation based on distance and cab type • Fleet management dashboard for administrators • Secure authentication using JWT tokens • Scalable architecture using the MERN stack These features make the system flexible, secure, and easy to extend with future features such as GPS tracking, online payments, and mobile app integration
4.	SocialImpact/CustomerSatisfaction	UCab improves transportation accessibility by making ride booking quick, transparent, and convenient. Benefits to Customers: • Easy and fast ride booking • Transparent fare estimation • Reduced waiting time • Reliable cab availability Benefits to Drivers and Fleet Owners: • Improved ride management • Increased ride opportunities • Centralized vehicle management Overall, the system contributes to digital transformation of local transportation services and improves urban mobility efficiency.
5.	BusinessModel(RevenueModel)	UCab can generate revenue through multiple channels: 1. Commission-Based Model A small percentage commission is charged on each ride. 2. Subscription Model Drivers or fleet owners may subscribe to premium plans for additional benefits. 3. Advertisement Model Local businesses can advertise on the platform. 4. Surge Pricing During high-demand periods fares may increase dynamically to balance supply and demand.
6.	Scalabilityofthe Solution	UCab is designed to support scalability using modern web technologies. • Backend APIs can be deployed on multiple servers to handle high traffic. • MongoDB supports distributed databases, replication, and sharding. • The application can be deployed on cloud platforms such as AWS, Google Cloud, or Azure. • The system architecture allows future integration with: - Mobile applications

		- Real-time GPS tracking - Online payment gateways - AI-based route optimization These capabilities ensure that the platform can grow as the number of users increases.